Final Paper

I.Title

Canteen Card System

II. Problem Statement

The reliance on cash transactions in school canteens shows significant problems like long wait times during break or lunch hours, errors made by humans, and the risk of students losing money. Furthermore, parents don't have a clear view of their children's spending habits, while canteen staff struggle to manage student debts due to disorganized records from manual record-keeping. This Canteen Card System fixes these real-world problems by providing a digital system. By removing the need for physical money and having automatic debt history and parental monitoring, this project shows a more secure financial way of transactions and encourages financial responsibility among students.

III. Project Overview And Relevance

The Canteen Card System is a multi-purpose financial management app that is built to align with the school cafeteria operations through distinct Student, Staff, and Parent options. The system uses Python for its logic, complex tasks such as role authentication, balance calculations, and transactions. It uses a strong database structure to maintain records of student balances, detailed transactions in histories, and debt reminders, ensuring data safety across different users. It is more than just a simple record-keeping; the project uses data analysis by allowing users to track spending over time, giving insights into canteen usage. This project serves as a demonstration of how software can transform manual tasks into a digital application.

IV. Objectives

The objective of this Canteen-Card System program is to allow users, such as students, parents, or staff and admins, to check the history and balance of their card at the Cafeteria if they have one via a program, allowing easier access.

Key Points/Objectives:

- Easier Access

- Less time and effort to check transactions and spending
- Consistent reminders whenever the balance is low

V. Methodology

The project was developed using the Python programming language. The system uses a simple database structure stored in JSON format to manage user data such as student information, balance, transactions, and purchases.

The program follows the CRUD process:

- **Create** – Users can register or be added to the system with a unique ID.
- **Read** – The system can display user information such as balance, debt, and transaction history.
- **Update** – Users can deposit money, purchase items, or update their account details.
- **Delete** – Certain records, such as transactions, may be removed if needed.

The system has different user roles:

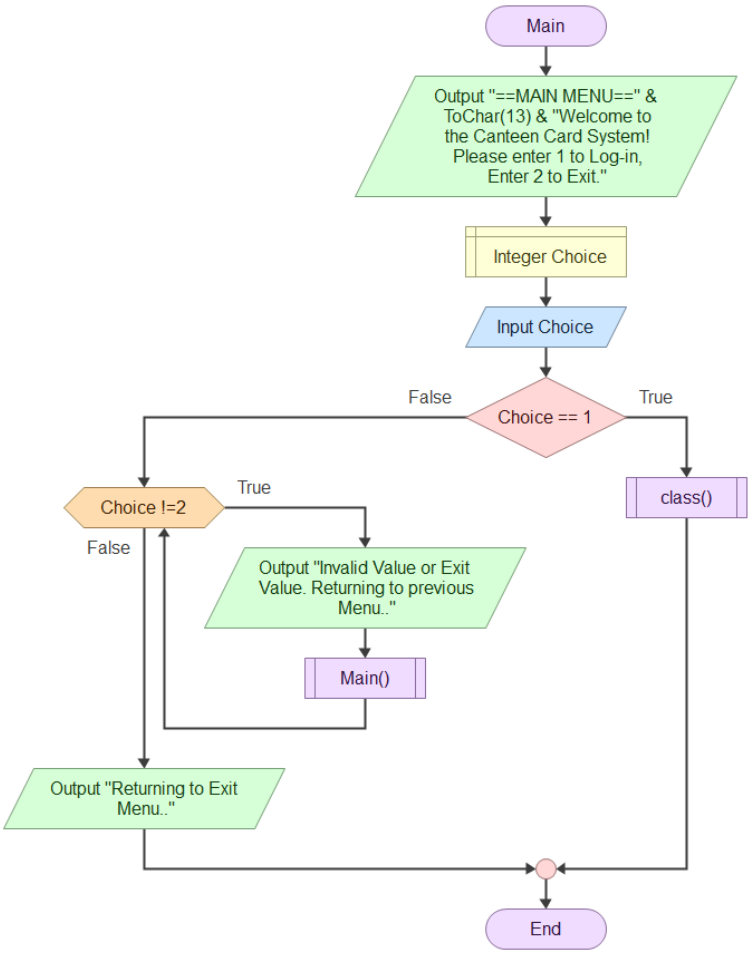
- **Students** – Can buy items, check balance, and view transactions.
- **Staff/Admin** – Has full access to manage all user data.
- **Parent** – Can view transaction history, balance, and debt.

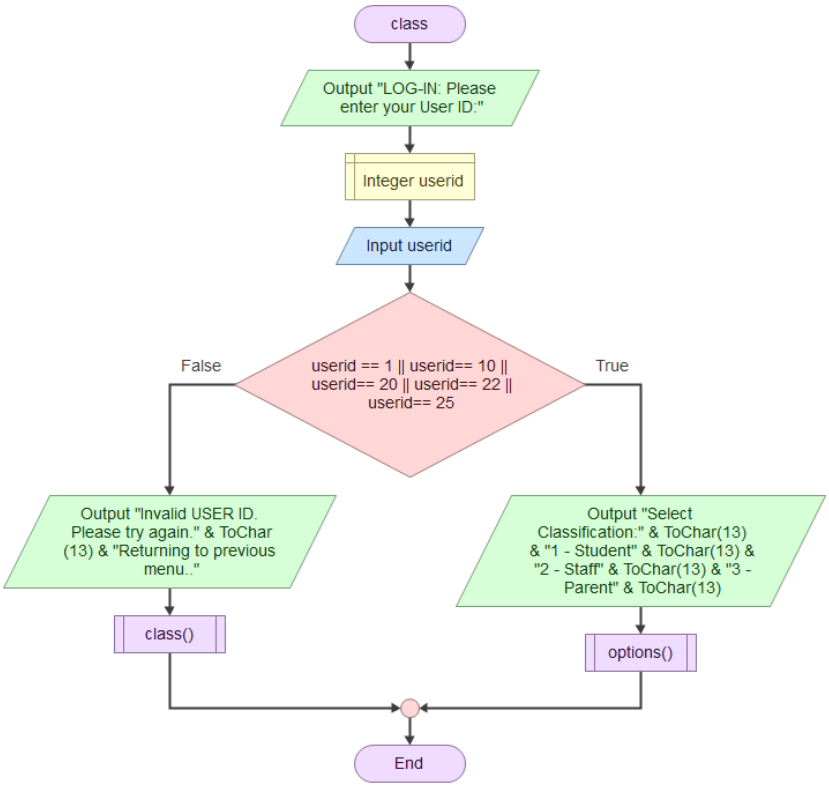
The program flow is controlled using functions and conditional statements. A menu system will guide users to select actions. Input validation is also applied to ensure correct data is entered.

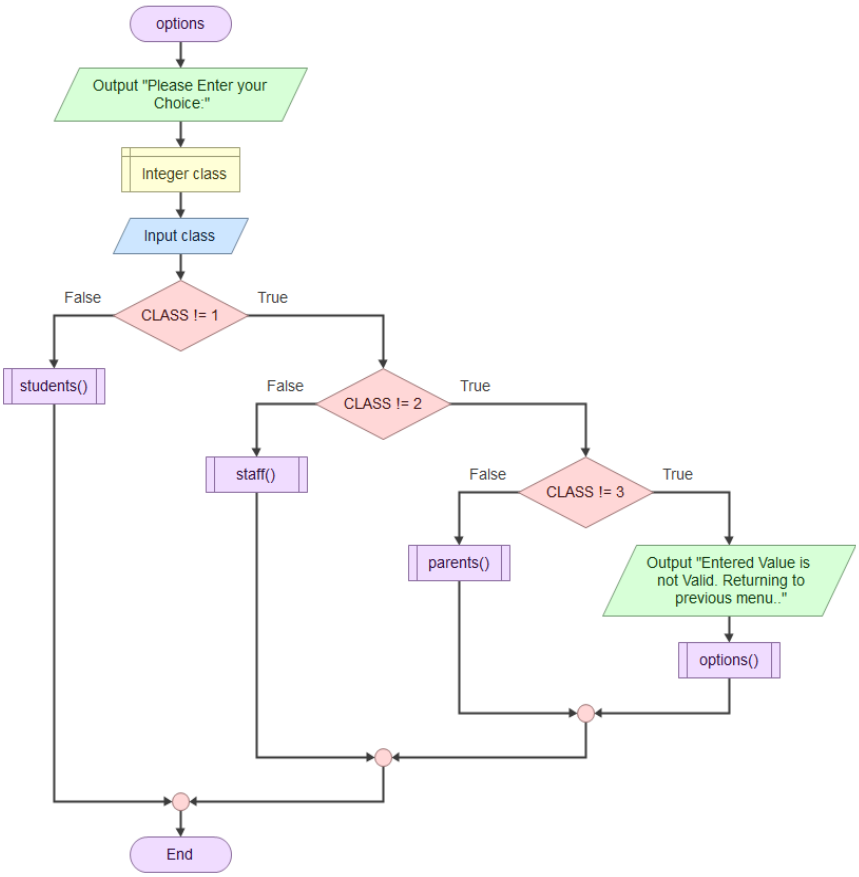
If Flask is used, the system can be turned into a simple web application where users interact through a browser instead of the console.

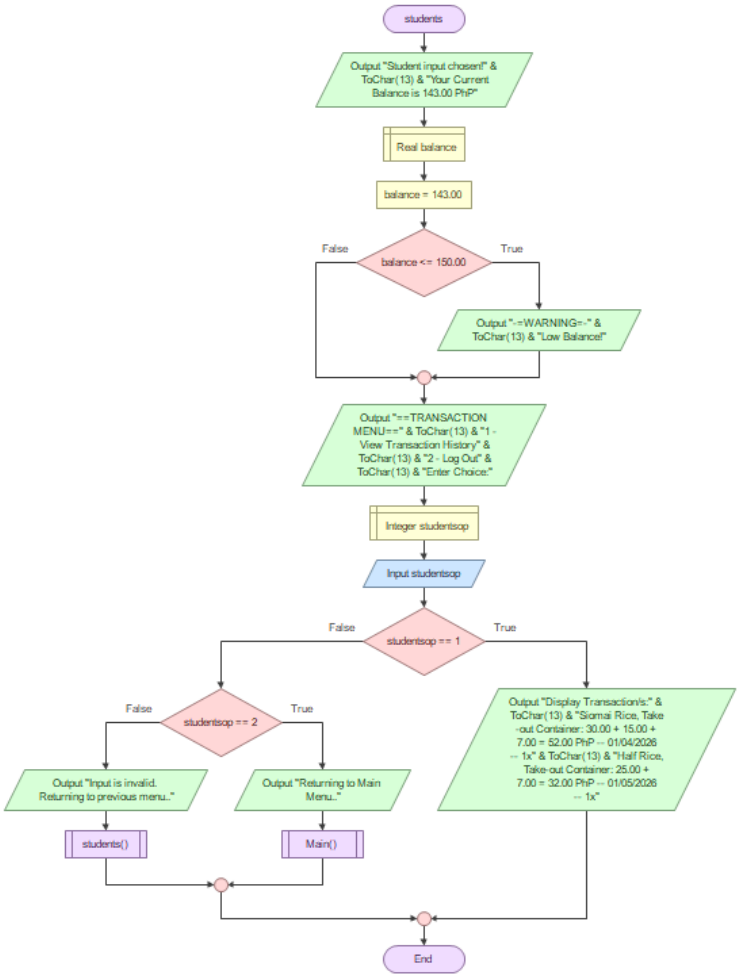
Basic data analysis may be included by tracking total spending, most purchased items, or user balances to help monitor usage of the system.

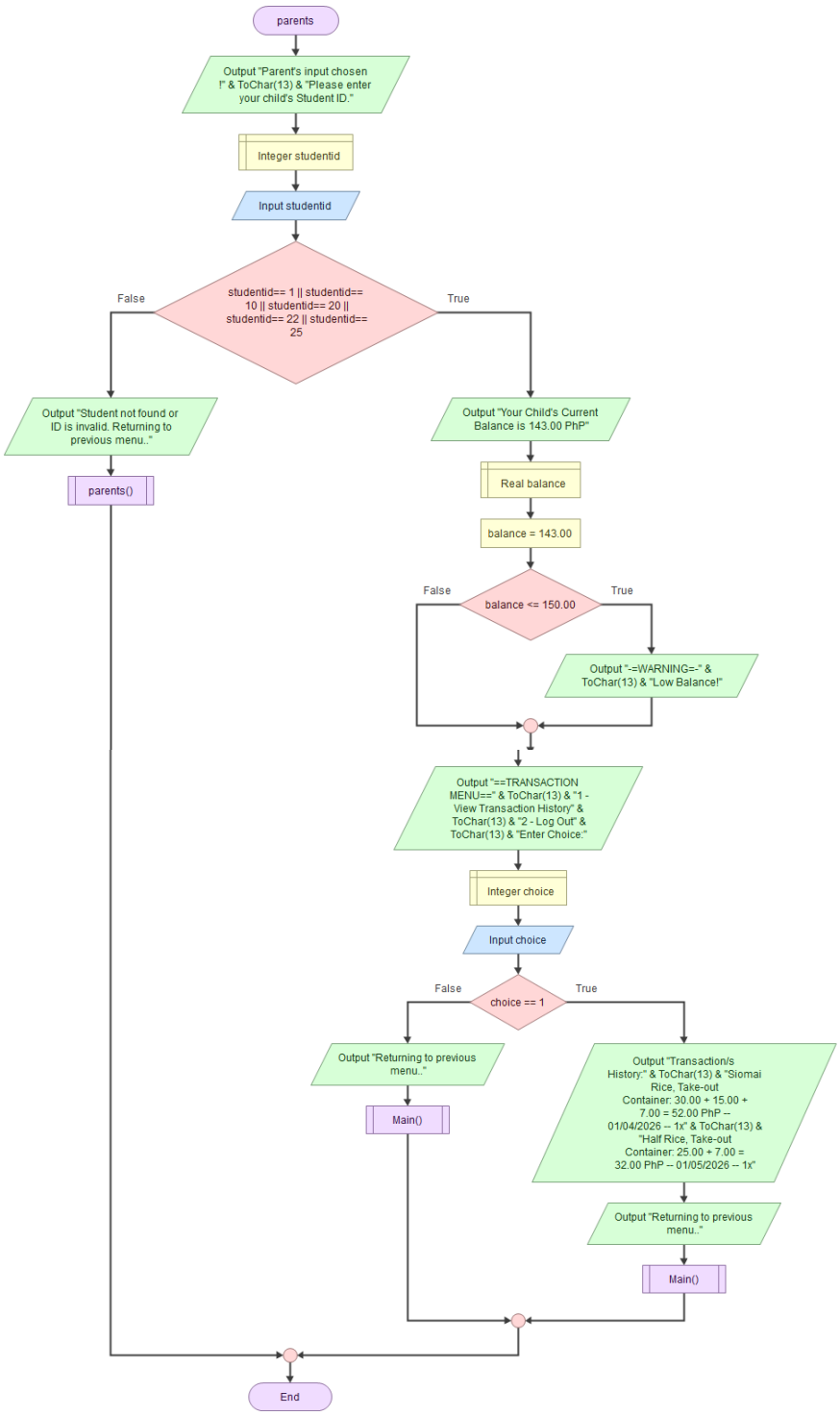
VI. Flowchart

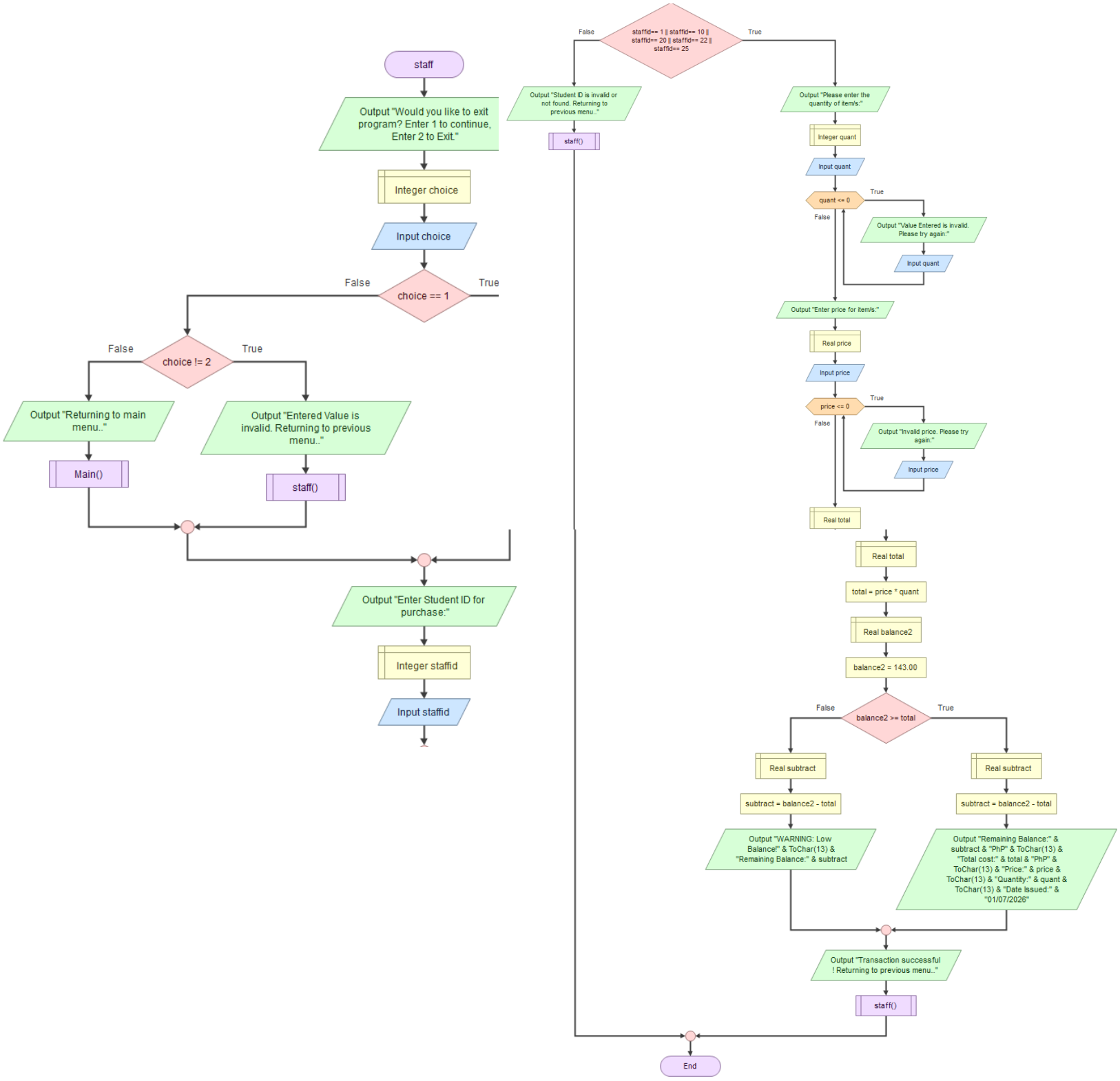









parents
Output "Parent's input chosen !" & ToChar(13) & "Please enter your child's Student ID."
Integer studentid
Input studentid
studentid== 1 || studentid== 10 || studentid== 20 || studentid== 22 || studentid== 25
False
True
Output "Student not found or ID is invalid. Returning to previous menu.."
parents()
Output "Your Child's Current Balance is 143.00 PhP"
Real balance
balance = 143.00
balance <= 150.00
False
True
Output "-=WARNING=-" & ToChar(13) & "Low Balance!"
Output "==TRANSACTION MENU==" & ToChar(13) & "1 - View Transaction History" & ToChar(13) & "2 - Log Out" & ToChar(13) & "Enter Choice:"
Integer choice
Input choice
choice == 1
False
True
Output "Returning to previous menu.."
Main()
Output "Transaction/s History:" & ToChar(13) & "Siomai Rice, Take-out Container: 30.00 + 15.00 + 7.00 = 52.00 PhP -- 01/04/2026 -- 1x" & ToChar(13) & "Half Rice, Take-out Container: 25.00 + 7.00 = 32.00 PhP -- 01/05/2026 -- 1x"
Output "Returning to previous menu.."
Main()
End


staff
Output "Would you like to exit program? Enter 1 to continue, Enter 2 to Exit."
Integer choice
Input choice
choice == 1
False
True
choice != 2
False
True
Output "Returning to main menu.."
Main()
Output "Entered Value is invalid. Returning to previous menu.."
staff()
Output "Enter Student ID for purchase:"
Integer staffid
Input staffid
staffid== 1 || staffid== 10 || staffid== 20 || staffid== 22 || staffid== 25
False
True
Output "Student ID is invalid or not found. Returning to previous menu.."
staff()
Output "Please enter the quantity of item/s:"
Integer quant
Input quant
quant <= 0
True
False
Output "Value Entered is invalid. Please try again:"
Input quant
Output "Enter price for item/s:"
Real price
Input price
price <= 0
True
False
Output "Invalid price. Please try again:"
Input price
Real total
Real total
total = price * quant
Real balance2
balance2 = 143.00
balance2 >= total
False
True
Real subtract
subtract = balance2 - total
Output "WARNING: Low Balance!" & ToChar(13) & "Remaining Balance:" & subtract
Real subtract
subtract = balance2 - total
Output "Remaining Balance:" & subtract & "PhP" & ToChar(13) & "Total cost:" & total & "PhP" & ToChar(13) & "Price:" & price & ToChar(13) & "Quantity:" & quant & ToChar(13) & "Date Issued:" & "01/07/2026"
Output "Transaction successful ! Returning to previous menu.."
staff()
End

VII. Program

VIII. Team Roles

Name	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Abadejos, Myles Deric D.C.	Proposed Solution, Problem Decomposition	Flowchart (Students)	Program Code (Students)	None
De Chavez, Kim Venidict B.	Problem Statement, Problem Decomposition	Project Decomposition	Program Code (Parents)	Project Overview /Statement and Conclusion
Manalo, Chelsea Margaret V.	Feasibility and Scope	Project Decomposition, Flowchart	Program Code (Overall + additional features)	PPT, Program
Mundo, Sachi Ysabelle M.	Problem Identification	Flowchart (Parents)	Program Code (Log-in)	Flowchart, Objectives, Readme File
Razon, Xedrick Loise A.	None	None	Program Code (Staff)	Methodology

IX. References